



# Managing Legacy CRC Operations Utilizing SAP ERP

Maintenance and  
Repair

Application

Component Life  
Management

Component  
Renewal (CRC)

MARC  
Management

|                                                  |   |
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## 1.0 Introduction

Long lead times and large costs associated with unplanned downtime of mine loading tools mandate that Component Rebuild Center (CRC) process owners have a clear line of sight to information required to manage work in progress associated with electric rope shovels, draglines and drills components being repaired at the CRC. SAP modules implemented in the legacy Bucyrus service centers did not come with turnkey metrics and reports geared towards CRC management, forcing individual regions to develop ad hoc reports.

It is possible to purchase software geared towards managing the constantly changing environment in a CRC. Typical software costs with installation are approximately \$75,000 and have an estimated total start to finish time of one year. The transition period associated with the transfer of Caterpillar service facilities to dealers presents a short window to improve performance within the current SAP enterprise resource planning software currently used at legacy Bucyrus service facilities prior to Dealer Day One.

This best practice discusses metrics and reports implemented at Caterpillar Canada's Edmonton CRC using existing reports from SAP. These metrics and reports help establish baselines for CRC performance and can be used to drive Continuous Improvement projects to increase efficiency, quality, and turnaround with Six Sigma, 5S, and CPS methodologies.

## 2.0 Best Practice Description

In order to facilitate an improvement initiative, it is critical that the project team be able to measure and record metrics and Key Performance Indicators (KPI's). These measurements are then used to analyze, improve and control performance.

Lack of metrics for CRC Work In Process (WIP) can lead to missed customer promise dates, poor customer experience, cost overruns and excessive delays from last labor to work order invoicing. It is common for WIP to stagnate if work orders requiring parts with long lead times are not followed up on a regular basis. Often these work orders are not started until the repair becomes a burning issue and result in overtime costs and damaged relations with the end customer.

In facilities where service exchange components are sold, it is critical to track cores owed from customers to ensure that they are returned in a timely manner. By establishing visual reports and metrics, process owners can follow up on potential problems before they escalate into machine down situations.

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The following reports and metrics were established at the Caterpillar Canada CRC:

## 2.1 WIP Dashboard Report:

The WIP Dashboard report presents a high level summary of all work orders.

| CATERPILLAR CANADA - EDMONTON CRC    |              |
|--------------------------------------|--------------|
| WORK IN PROGRESS – DASHBOARD         | 9/30/2012    |
| Service work order margin %          | 41%          |
| Average Promise Date Adherence - %   | 59%          |
| Average Total Days Order Open        | 181          |
| Total WIP - \$                       | \$ 8,084,817 |
| Total Count Of Work Orders           | 417          |
| Average Last Labor to Invoice (Days) | 57           |
| Labor Utilization Metrics            |              |
| Total Productive Hours               | 24136        |
| Total Overhead Hours                 | 5461         |
| Total Labor Utilization              | 82%          |

Table 1: WIP Dashboard Report

Metrics included in this report are:

### 2.1.1 Service work order margin %

Process owners can avoid losses by reviewing margin associated with orders in WIP. Any work orders at risk of not meeting margin targets can be managed to mitigate losses.

### 2.1.2 Promise Date Adherence

Work orders at risk of not meeting the date promised to customers are flagged for process owner action. Orders at risk of missing the promised delivery date can be identified and addressed.

### 2.1.3 Total Days Order Open

Orders open for an excessive period of time tie up WIP funds and are indicators of process issues. Monitoring total days since order creation can help identify these issues.

### 2.1.4 Total Days since Last Labor to Invoice (LLI)

Excessive LLI delays the timing of cash flows, negatively impacting a company's financial position. Reducing this metric results in improved profitability by eliminating WIP holding charges and increases customer satisfaction via efficient invoicing practices. Benchmarks for LLI range from 6-10 days. The appropriate goal will vary based on factors surrounding obtaining customer purchase order numbers, parts/core return processes, report reviews, and many others.

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### 2.1.5 Labor Utilization Report:

This metric tracks labor utilization at the CRC. By tracking labor utilization, process owners can address gaps and bottleneck in planning and scheduling activities at the CRC. Unutilized labor is often the result of the Caterpillar 8 wastes. Identifying and eliminating these wastes will result in and improvement in the labor utilization metric.

### 2.2 Service Exchange Core Tracker:

Delays associated with the return of cores from customer sites can result in longer repair lead times and in some cases, lost cores. This report presents a count of cores owed by customer and as well as potential core charges associated with the outstanding cores

| Canada Service Exchange - Core Tracker Summary |            |                 |              |              |                 |
|------------------------------------------------|------------|-----------------|--------------|--------------|-----------------|
| Account Manager                                | Customer   | Cores > 60 days | Core Charge  | 30 - 60 days | Core \$ 30 - 60 |
| AM 1                                           | CUSTOMER 1 | 44              | \$ 937,331   | 22           | \$ 339,864      |
| AM 2                                           | CUSTOMER 2 | 46              | \$ 1,572,937 | 5            | \$ 40,507       |
| AM 3                                           | CUSTOMER 3 | 3               | \$ 78,618.00 | 2            | \$ 44,057       |
| AM 4                                           | CUSTOMER 4 | 2               | \$ 43,807    | 11           | \$ 669,096      |

Table 2: Service Exchange Core Tracker

### 2.3 Service Exchange (SX) Dashboard:

This report presents a snapshot of current levels of Inventory, WIP and outstanding cores by part number, allowing the process owner to ensure that the correct inventory level is available to serve service exchange customers.

| Canada: Service Exchange Inventory Dashboard |                                          |           |            |     |                    |               |       |
|----------------------------------------------|------------------------------------------|-----------|------------|-----|--------------------|---------------|-------|
| PART                                         | PART DESCRIPTION                         | INVENTORY | CORES OWED | WIP | TOTAL SX INVENTORY | STOCKING PLAN | DELTA |
| 1524416-SR                                   | DUMP CYLINDER REBUILD                    | 1         | 0          | 0   | 1                  | 1             | 0     |
| 1521221-SR                                   | SWING PUMP REBUILD                       | 0         | 0          | 0   | 0                  | 1             | -1    |
| 1521197-SR                                   | VARIABLE PUMP                            | 1         | 0          | 0   | 1                  | 1             | 0     |
| 1521134-SR                                   | HYDRAULIC MOTOR REBUILD                  | 0         | 1          | 0   | 1                  | 1             | 0     |
| 1471865-SR                                   | DRIVE, PUMP GEARBOX                      | 0         | 0          | 0   | 0                  | 1             | -1    |
| 1467010-SR                                   | DUMP CYLINDER, REBUILD                   | 3         | 0          | 0   | 3                  | 2             | 1     |
| 1461462-SR                                   | REM -TPI 1461462-SR RH170 TRAVEL MOTOR B | 0         | 0          | 1   | 1                  | 1             | 0     |
| 1459401-SR                                   | PUMP HYD COOLING FAN RH200, REBUILD      | 1         | 0          | 0   | 1                  | 1             | 0     |
| 1459392-SR                                   | REM 1459392-SR MAIN VARIABLE PUMP RGA 11 | 0         | 0          | 0   | 0                  | 5             | -5    |
| 1459391-SR                                   | PUMP, MAIN "OLD CONFIGURATION"           | 0         | 1          | 0   | 1                  | 2             | -1    |
| 1456272-SR                                   | REM - 1456272-SR Swing Pump RGA93362-1   | 0         | 0          | 0   | 0                  | 1             | -1    |
| 1455174-SR                                   | REM 1455174-SR VARIABLE PUMP BOSCH RGA 1 | 6         | 0          | 0   | 6                  | 1             | 5     |
| 1453560-SR                                   | REM 1453560-SR ROTOR NEW STYLE INOTEC W/ | 2         | 0          | 0   | 2                  | 2             | 0     |
| 1453559-SR                                   | ROTOR, OLD STYLE                         | 1         | 0          | 0   | 1                  | 2             | -1    |
| 1453555-SR                                   | DRIVE, NEW STYLE PUMP DRIVE              | 1         | 1          | 0   | 2                  | 1             | 1     |
| 1453554-SR                                   | DRIVE, FINAL                             | 1         | 0          | 1   | 2                  | 1             | 1     |
| 1453549-SR                                   | CYLINDER, BUCKET CURL                    | 1         | 0          | 0   | 1                  | 2             | -1    |
| 1453544-SR                                   | CYLINDER, FS STICK                       | 1         | 0          | 0   | 1                  | 2             | -1    |
| 1451867-SR                                   | REM – TPI – 1451867-SR S54 SWING TRANSMI | 0         | 1          | 0   | 1                  | 1             | 0     |
| 1451773-SR                                   | Trans swing old style 1451773-SR         | 0         | 1          | 0   | 1                  | 7             | -6    |
| 1451680-SR                                   | DRIVE, NEW STYLE FINAL                   | 4         | 0          | 0   | 3                  | 2             | 1     |

Table 3: Service Exchange Inventory Dashboard

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|-----------------------------------------------------------------------------------------------------------------|-------------------|--------------|--------------------------|
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### 3.0 Implementation Steps

The high level process steps involved with implementation of these metrics are:

- Establish work in progress management team
- Identify and assign ownership of WIP reporting
- Create reports by combining standard SAP reports utilizing feedback from process owners
- Measure and communicate metrics
- Document benefits
- Identify Continuous Improvement (CI) projects based on collected metrics

The SAP transaction codes required to compile the metrics are:

- CATS\_DA – *Display Working Times*
- CJI3 – *Display Actual Cost Line Items*
- IWBK – *Material Availability Information*
- IW37 – *Change Operations*
- IW38 – *Change Planned Maintenance (PM) Orders*
- ZSD033 – *Sales Order Report*
- Z\_DEV\_51000056 – *Invoiced Sales Report*
- Z\_CORES – *Cores Owed By Customer*
- ZCUST\_CONSIGN – *Consignment inventory*
- Z\_PLANT\_INVENTORY – *Plant Inventory*

Output from these sessions is used to create the required metrics, see attached spreadsheets for sample calculations of each metric from the standard SAP Reports

#### 3.1 WIP Dashboard Report:

This report is run from SAP transaction code IW38 (See attached example). The default report layout will have to be modified to include all the required fields. Layouts can be saved by the user for future use.

##### 3.1.1 Service work order margin %

Compare the actual cost for work orders on IW38 by totaling all costs for parent and child orders from CJI3. CJI3 will have to be run with all order numbers from IW38 entered into the order field. Compare to the contract value for associated sales order in ZSD033 for margin calculation.

##### 3.1.2 Promise Date Adherence

Generate by comparing the sales order due date in ZSD033 to the current date. Cross reference with work order from IW38 for ease of recognition.

##### 3.1.3 Total Days Order Open

Calculate total days that each order has been open order by comparing the current date to the order creation date on IW38

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### 3.2 Total Days since Last Labor to Invoice (LLI)

Determine total days LLI by comparing the last labor charged data from CJI3 for each service document invoice creation date on invoice report.. Obtain service orders for all invoiced service documents from IW38 (include completed and historical orders). Use service orders numbers to run CJI3 for last labor dates (It is important to only include labor lines). Compare last labor date to invoice billing date to determine LLI. In the case of multiple invoice dates, eliminate duplicates and only include last invoice date.

### 3.3 Labor Utilization Report:

Use a combination of data from CJI3 and CATS\_DA to determine labor utilization. By using CATS\_DA, it is possible to determine all labor charged to work orders for a given time period. This value can then be combined with CJI3 data for overhead accounts to determine utilization. Note: It will be necessary to determine local GL code for indirect labor. Consult with costs analyst in finance for GL code.

### 3.4 Service Exchange Core Tracker:

This report is derived by running transaction code Z\_Cores. The standard report from SAP will show all service exchange components and will indicate exchange posted date for components that have been shipped to the customer. The report can be further formatted to include account representative information and core charge information that is maintained in a manual table.

### 3.5 Service Exchange Dashboard:

Use plant specific out put from Z\_Plant\_Inventory, and WIP level from IW38. You will also need Sales Backlog to populate part numbers associated with IW38 orders

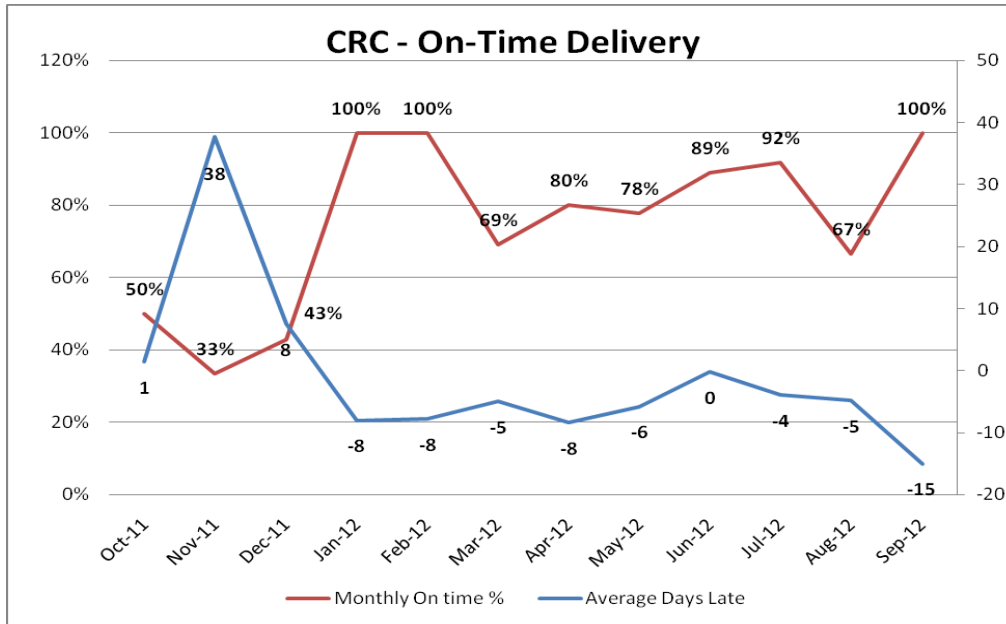
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#### 4.0 Benefits

Defined KPI's provide the process owner a clear line of sight to service operation performance. These metrics can then be used for future continuous improvement projects.

The reports generated from this project were used to manage a component repair contract with a customer resulting in the CRC meeting the customer's on-time delivery expectations and achieving a 3 month rolling average improvement from 42% to 86%.



In addition, along with some 5S activities, a reduction in active WIP resulted in improved utilization of shop floor space and a better work environment, as indicated in Figure 1 and 2 below. Note work bays and work in process parts staging are much better organized, improving efficiency and work flow.

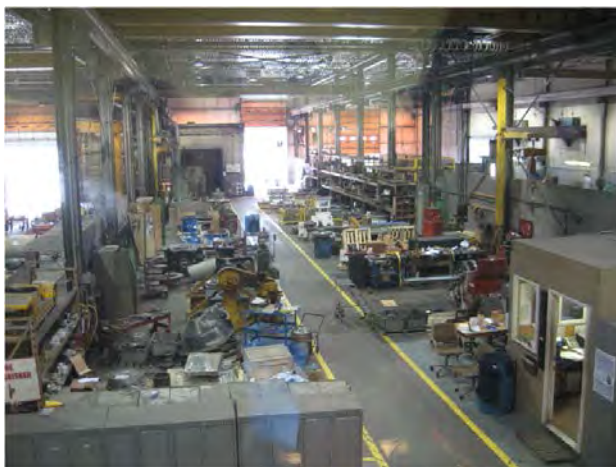


Figure 1: Shop Floor Before



Figure 2: Shop Floor After

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Finally, these metrics can be used to monitor health of WIP for transition from Caterpillar to Dealers and to resolve problems prior to Dealer Day One transitions in facilities utilizing SAP.

## 5.0 Resources Required

Personnel time to create and communicate reports

## 6.0 Supporting Attachments / References

- WIP Dashboard
- LLI Calculation
- Labor Utilization Calculation
- Service Exchange Core Tracker
- Service Exchange Dashboard

## 7.0 Related Best Practices

- 0906-4.5-1013 – CRC Component Turnaround
- 0208-4.2-1114 – CRC Planning & Scheduling Systems
- 0107-4.5-1055 – Managing Exchange Component Inventory Reduces Costs
- 0307-4.5-1069 – Improving the Last Labor to Work Order Invoicing (LLI) Process

## 8.0 Acknowledgements

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## 9.0 Appendix

### 9.1 CATS\_DA – Display Working Times

This report should be run for the period that the labor analysis is required for. Company code is dependent on local company code.

Microsoft Excel - LLI Calculation

File Edit View Insert Format Tools Data Window Help

Program Edit Goto System Help

Display Working Times

OrgStructure Search Help &000000000000004

Period

Reporting Period Other Period 09/01/2012 - 09/30/2012

Selection Criteria

Personnel Number

Employment Status

Company Code 1300

Cost Center

Selection of Time Sheet

Basic Data

|                      |    |    |    |
|----------------------|----|----|----|
| Activity Type        |    | to |    |
| Task level           |    | to |    |
| Task component       |    | to |    |
| Activity Type        |    | to |    |
| Stat. key figure     |    | to |    |
| Att./Absence type    |    | to |    |
| Wage Type            |    | to |    |
| Display Unit/Measure |    | to |    |
| Number (unit)        |    | to |    |
| Processing status    | 10 | to | 40 |
| Short Text           |    | to |    |

Receiver account assgmt

|                   |  |    |  |
|-------------------|--|----|--|
| Rec. sales order  |  | to |  |
| RecSalesOrd. item |  | to |  |
| Receiver Order    |  | to |  |

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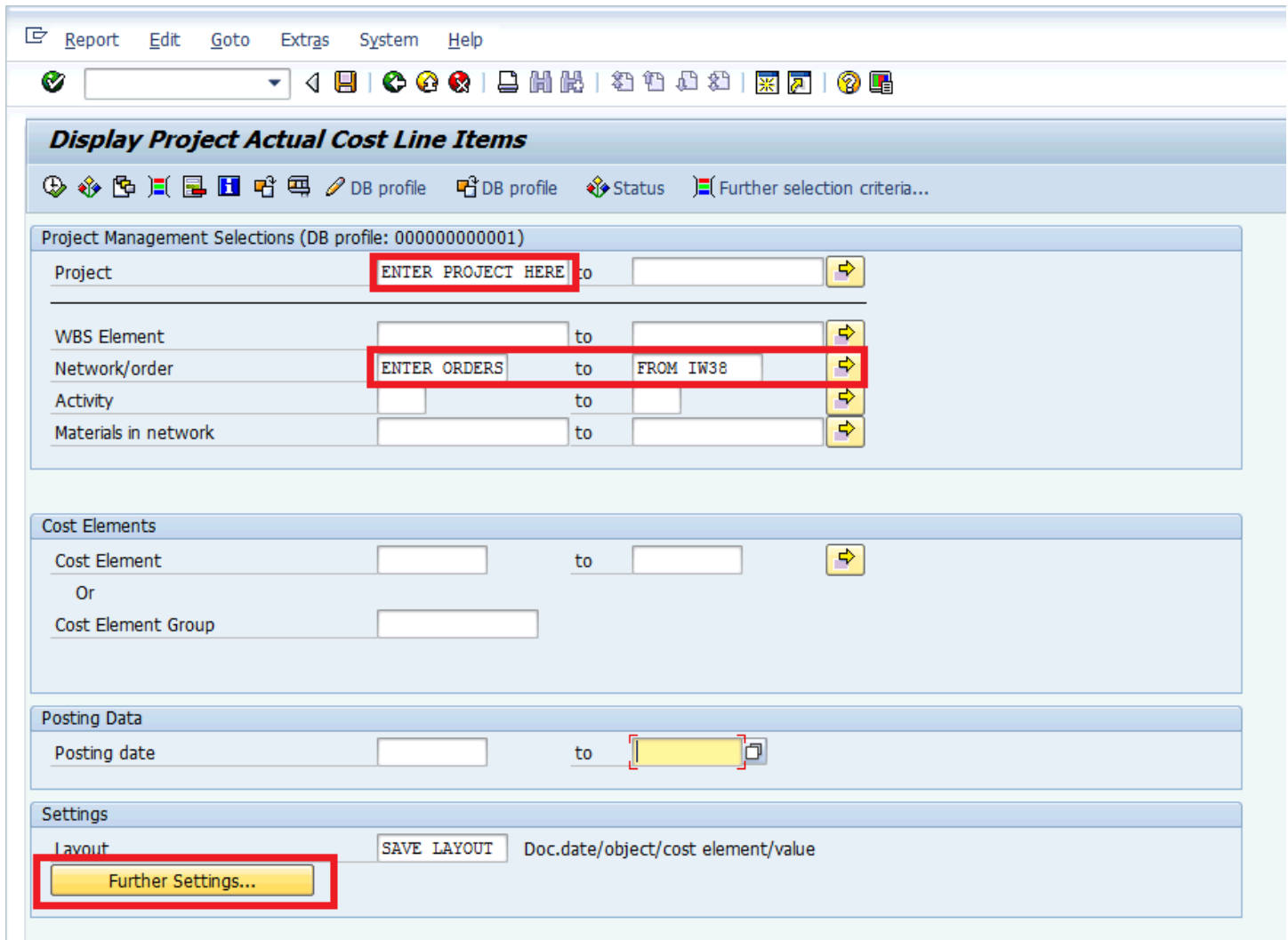
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## 9.2 CJ13 – Display Actual Cost Line Items

- Project is required only when calculating overhead costs for labor utilization.
- Network/order should be filled in from IW38 work orders.
- Ensure that  is used to copy work order list from IW38. Also note that  should be used and not “Ctrl + V” to past work order list.
- Update further settings to “999,999,999”
- For controlling area see financial rep
- Profile is “000000000001”



**Display Project Actual Cost Line Items**

Project Management Selections (DB profile: 000000000001)

Project  to

WBS Element  to

Network/order  to

Activity  to

Materials in network  to

Cost Elements

Cost Element  to

Or

Cost Element Group

Posting Data

Posting date  to

Settings

Layout  Doc.date/object/cost element/value

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### 9.3 IWBK – Material Availability Information

- Order should be filled in from IW38 work orders.
- Ensure that  is used to copy work order list from IW38. Also note that  should be used and not “Ctrl + V” to paste work order list.

Program Edit Goto System Help

#### Material Availability Information

Order status

☒ Outst ☒ In procss ☐ Completed ☐ Histor.

|                            |              |    |           |                                                                                      |                                                                                      |
|----------------------------|--------------|----|-----------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Order                      | ENTER ORDERS | to | FROM IW38 |     |    |
| Order type                 |              | to |           |    |    |
| Functional location        |              | to |           |    |    |
| Equipment                  |              | to |           |    |    |
| Material                   |              | to |           |    |   |
| Serial number              |              | to |           |  |  |
| Notification               |              | to |           |  |  |
| Maintenance planning plant |              | to |           |  |  |
| Entered by                 |              | to |           |  |  |
| Date entered               |              | to |           |  |  |
| Basic start date           |              | to |           |  |  |
| Basic finish date          |              | to |           |  |  |
| Scheduled start            |              | to |           |  |  |
| Scheduled finish           |              | to |           |  |  |
| Actual start date          |              | to |           |  |  |
| Material (reservation)     |              | to |           |  |  |
| Scheduled release date     |              | to |           |  |  |
| Output variant             |              |    |           |                                                                                      |                                                                                      |

Print

☐ Print immediately

☐ No selection information

☐ No column headings

☐ No new page

☐ No list information

Number of footer lines

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#### 9.4 IW37 – Change Operations

- Order should be filled in from IW38 work orders.
- Ensure that  is used to copy work order list from IW38. Also note that  should be used and not “Ctrl + V” to past work order list.

Program Edit Goto System Help



### Change Operations: Selection of Order Operations



Operations

☒ Current ☐ Historical

Order operation selection

|                                                       |                                                                                   |    |  |                                                                                       |
|-------------------------------------------------------|-----------------------------------------------------------------------------------|----|--|---------------------------------------------------------------------------------------|
| Work center                                           |                                                                                   | to |  |    |
| Plant                                                 |  | to |  |    |
| Order                                                 |                                                                                   | to |  |    |
| Order Type                                            |                                                                                   | to |  |    |
| Reference date                                        |                                                                                   | to |  |   |
| Operation/Activity                                    |                                                                                   | to |  |  |
| Operation short text                                  |                                                                                   | to |  |  |
| Functional location                                   |                                                                                   | to |  |  |
| Equipment                                             |                                                                                   | to |  |  |
| Material                                              |                                                                                   | to |  |  |
| Serial number                                         |                                                                                   | to |  |  |
| Sort field                                            |                                                                                   | to |  |  |
| Planner group                                         |                                                                                   | to |  |  |
| Planning plant                                        |                                                                                   | to |  |  |
| Confirmation                                          |                                                                                   | to |  |  |
| Control key                                           |                                                                                   | to |  |  |
| Status inclusive                                      |                                                                                   | to |  |  |
| Status exclusive                                      |                                                                                   | to |  |  |
| Assembly                                              |                                                                                   | to |  |  |
| Standard text key                                     |                                                                                   | to |  |  |
| System Condition                                      |                                                                                   | to |  |  |
| <input checked="" type="checkbox"/> Service/PM orders |                                                                                   |    |  |                                                                                       |

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## 9.5 IW38 – Change PM Orders

- Report should be run without any selections (run everything blank) to ensure that all possible orders are captured

Viewing Robert's desktop

Program Edit Goto System Help

Change PM Orders: Selection of Orders

Settlement Receivers PRT

Order status

☒ Outstanding ☒ In process ☐ Completed ☐ Historical Sel.profil  Addr. ✖

Order selection

|                      |                      |    |                      |   |
|----------------------|----------------------|----|----------------------|---|
| Order                | <input type="text"/> | to | <input type="text"/> | ➡ |
| Order Type           | <input type="text"/> | to | <input type="text"/> | ➡ |
| Functional location  | <input type="text"/> | to | <input type="text"/> | ➡ |
| Equipment            | <input type="text"/> | to | <input type="text"/> | ➡ |
| Material             | <input type="text"/> | to | <input type="text"/> | ➡ |
| Serial number        | <input type="text"/> | to | <input type="text"/> | ➡ |
| Addit. device data   | <input type="text"/> | to | <input type="text"/> | ➡ |
| Notification         | <input type="text"/> | to | <input type="text"/> | ➡ |
| Main work center     | <input type="text"/> | to | <input type="text"/> | ➡ |
| Plant for WorkCenter | <input type="text"/> | to | <input type="text"/> | ➡ |
| Period               | <input type="text"/> | to | <input type="text"/> |   |
| Partners             | <input type="text"/> |    |                      |   |
| Currency             | <input type="text"/> |    |                      |   |

General Data/Administrative Data

☐ incl.object list

|                |                      |    |                      |   |
|----------------|----------------------|----|----------------------|---|
| Leading order  | <input type="text"/> | to | <input type="text"/> | ➡ |
| Superior order | <input type="text"/> | to | <input type="text"/> | ➡ |
| Planning plant | <input type="text"/> | to | <input type="text"/> | ➡ |
| Priority       | <input type="text"/> | to | <input type="text"/> | ➡ |
| Entered by     | <input type="text"/> | to | <input type="text"/> | ➡ |

SAP

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## 9.6 ZSD033 – Sales Order Report

- Sales organization is dependent on local organization
- Distribution channel and division should be entire range from dropdown menu
- Plant is range should include all applicable plants
- Sales order should be copied from IW38 for all applicable service sales orders

Program Edit Goto System Help

Open Sales Order Report

Selection for Sales Area

|                      |                                     |    |                      |  |
|----------------------|-------------------------------------|----|----------------------|--|
| Sales Organization   | <input checked="" type="checkbox"/> | to | <input type="text"/> |  |
| Distribution Channel | <input checked="" type="checkbox"/> | to | <input type="text"/> |  |
| Division             | <input checked="" type="checkbox"/> | to | <input type="text"/> |  |

|                     |                                     |    |                      |  |
|---------------------|-------------------------------------|----|----------------------|--|
| Plant               | <input checked="" type="checkbox"/> | to | <input type="text"/> |  |
| Sales Order         | <input type="text"/>                | to | <input type="text"/> |  |
| Material Number     | <input type="text"/>                | to | <input type="text"/> |  |
| Created by          | <input type="text"/>                | to | <input type="text"/> |  |
| Sold-to party       | <input type="text"/>                | to | <input type="text"/> |  |
| Created on          | <input type="text"/>                | to | <input type="text"/> |  |
| Delivery Date       | <input type="text"/>                | to | <input type="text"/> |  |
| Sales Document Type | <input type="text"/>                | to | <input type="text"/> |  |
| Order reason        | <input type="text"/>                | to | <input type="text"/> |  |
| Delivery Priority   | <input type="text"/>                | to | <input type="text"/> |  |
| Your Reference      | <input type="text"/>                | to | <input type="text"/> |  |
| WBS Element         | <input type="text"/>                | to | <input type="text"/> |  |
| Overall status      | <input type="text"/>                | to | <input type="text"/> |  |

Output List Options

☒ Order Report  
☐ Order Complete  
☐ Open Order

Addition Selection for Open Orders

☐ With Resources Detail  
☐ Include component requirements

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## 9.7 Z\_DEV\_51000056 – Invoiced Sales Report

- Posting period and fiscal year is for period being analyzed
- Sales organization is local sales organization
- Sales document range should be large enough to include all possible service sales documents

Program Edit Goto System Help

**Invoiced Sales Report For Make-to-Order**

**General data selections**

|                  |      |    |      |  |
|------------------|------|----|------|--|
| Posting Period   | 1    | to | 9    |  |
| Valuation Class  |      | to |      |  |
| Fiscal Year      | 2012 | to | 2012 |  |
| Sales Order Type |      | to |      |  |

**Report-specific selections**

|                           |           |    |           |  |
|---------------------------|-----------|----|-----------|--|
| Account assignment group  |           | to |           |  |
| Status for transfer to ac |           | to |           |  |
| Sales Office              |           | to |           |  |
| Product hierarchy         |           | to |           |  |
| Material Group            |           | to |           |  |
| SD document category      |           | to |           |  |
| Sold-to party             |           | to |           |  |
| Billing category          |           | to |           |  |
| Billing Type              |           | to |           |  |
| Sales Organization        | 1300      | to |           |  |
| Sales Document            | 500000000 | to | 600000000 |  |
| Distribution Channel      |           | to |           |  |
| Material Number           |           | to |           |  |
| Billing Document          |           | to |           |  |
| Billing date for billing  |           | to |           |  |
| Document number of the re |           | to |           |  |
| Delivery Date             |           | to |           |  |
| Incoterms (Part 1)        |           | to |           |  |
| Work Breakdown Structure  |           | to |           |  |
| Profit Center             |           | to |           |  |

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## 9.8 Z\_CORES – Cores Owed By Customer

- Sales organization is local sales organization
- Plant is local plant

Program Edit Goto System Help

1300

### Cores Owed from Customer

Report-specific selections

|                      |      |    |   |   |
|----------------------|------|----|---|---|
| Sales Organization   | 1300 | to |   | → |
| Distribution Channel |      | to |   | → |
| Plant                |      | to |   | → |
| Storage Location     |      | to |   | → |
| Division             |      | to |   | → |
| Ship-To              |      | to |   | → |
| Delivery status      | A    | to | B | → |
| Order Reason         |      | to |   | → |
| Sales Document       |      | to |   | → |
| Creation Date        |      | to |   | → |
| Service Product      |      | to |   | → |
| Reason for Rejection | =    | to |   | → |
| Order Reason         |      | to |   | → |
| Sales Document Item  |      | to |   | → |
| Returns Item         | X    | to |   | → |
| Partner Function     | SH   | to |   | → |

Output specification

Layout

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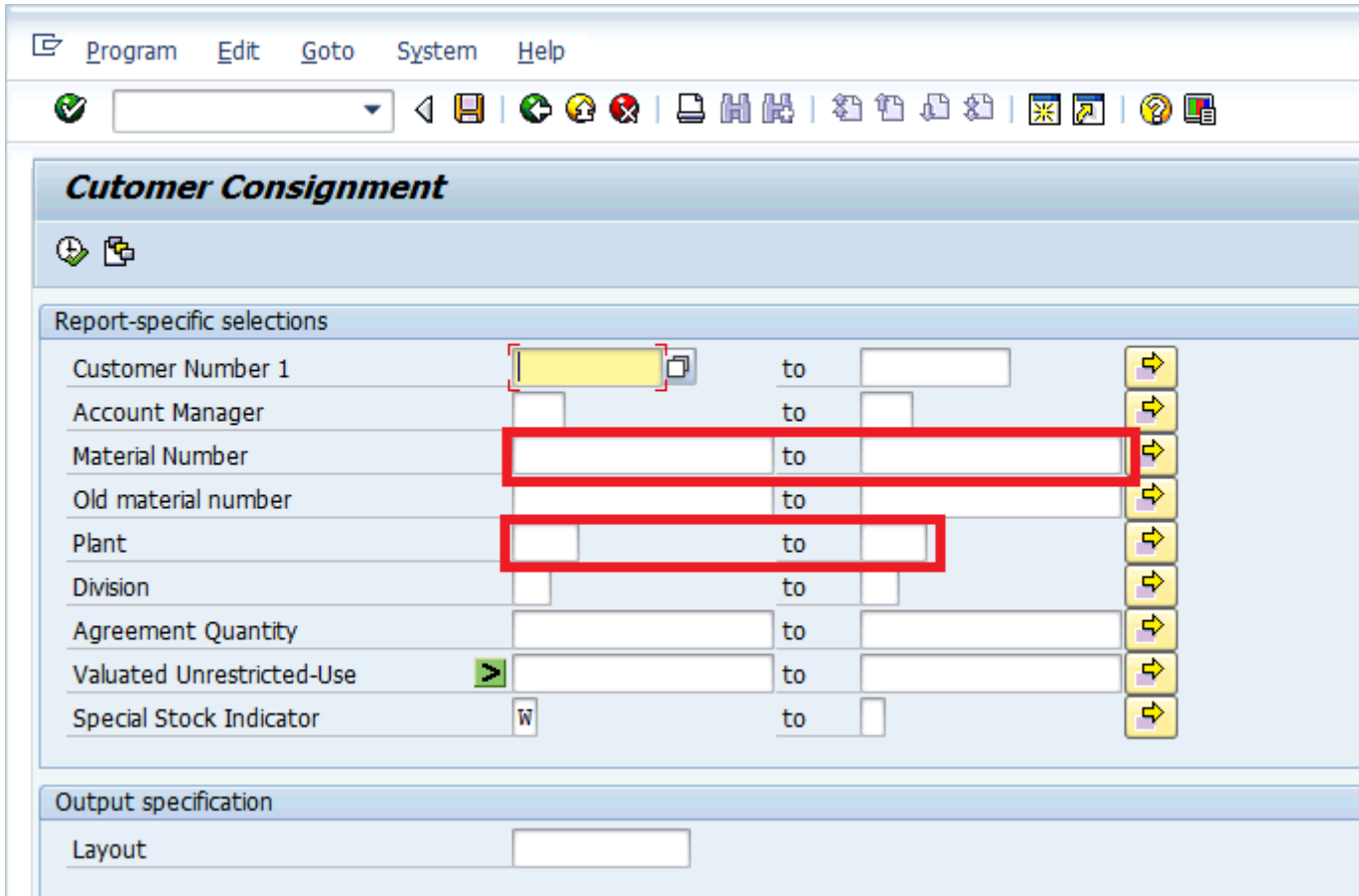
CHG  
NO  
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NUMBER  
1210-4.04-1287



### 9.9 ZCUST\_CONSIGN – Consignment inventory

- Material number range should be derived from local Service exchange stocking plan. If one does not exist, material number “\*SR\*” can be used to include all service part numbers in stock
- Ensure that  is used to copy part number list. Also note that  should be used and not “Ctrl + V” to past work order list.



**Customer Consignment**

Report-specific selections

|                           |                                                                                     |    |                                                                                       |
|---------------------------|-------------------------------------------------------------------------------------|----|---------------------------------------------------------------------------------------|
| Customer Number 1         |    | to |    |
| Account Manager           |                                                                                     | to |    |
| Material Number           |  | to |    |
| Old material number       |                                                                                     | to |    |
| Plant                     |                                                                                     | to |    |
| Division                  |                                                                                     | to |   |
| Agreement Quantity        |                                                                                     | to |  |
| Valuated Unrestricted-Use |  | to |  |
| Special Stock Indicator   | W                                                                                   | to |  |

Output specification

Layout

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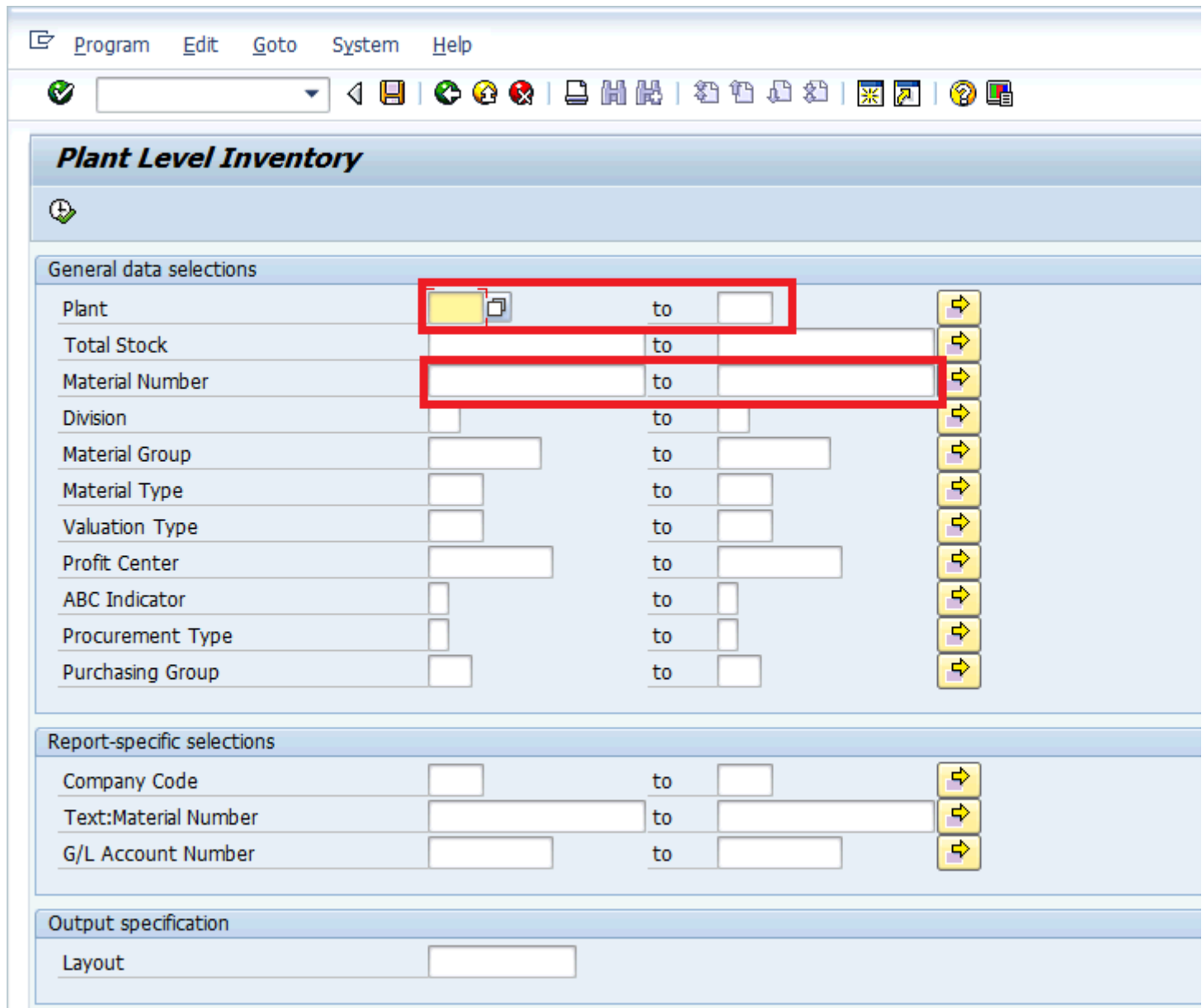
DATE  
2/20/2015

CHG  
NO  
00

NUMBER  
1210-4.04-1287

**9.10 Z\_PLANT\_INVENTORY – Plant Inventory**

- Plant is local plant number
- Material number range should be derived from local Service exchange stocking plan. If one does not exist, material number “\*SR\*” can be used to include all service part numbers in stock
- Ensure that  is used to copy part number list. Also note that  should be used and not “Ctrl + V” to past work order list.



**Plant Level Inventory**

General data selections

|                  |                                                                                   |    |                                                                                       |
|------------------|-----------------------------------------------------------------------------------|----|---------------------------------------------------------------------------------------|
| Plant            |  | to |    |
| Total Stock      |                                                                                   | to |    |
| Material Number  |  | to |    |
| Division         |                                                                                   | to |    |
| Material Group   |                                                                                   | to |   |
| Material Type    |                                                                                   | to |  |
| Valuation Type   |                                                                                   | to |  |
| Profit Center    |                                                                                   | to |  |
| ABC Indicator    |                                                                                   | to |  |
| Procurement Type |                                                                                   | to |  |
| Purchasing Group |                                                                                   | to |  |

Report-specific selections

|                      |  |    |                                                                                       |
|----------------------|--|----|---------------------------------------------------------------------------------------|
| Company Code         |  | to |  |
| Text:Material Number |  | to |  |
| G/L Account Number   |  | to |  |

Output specification

|        |  |
|--------|--|
| Layout |  |
|--------|--|

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